Networking standards

Networking standards define the rules for data communications that are needed for interoperability of networking technologies and processes.

Standards are the set of rules for data communication that are needed for exchange of information among devices. It is important to follow Standards which are created by various Standard Organization like ISO ,IEEE , ANSI etc.

During data communication, a number of standards may be used simultaneously at the different layers.

Standards Organizations

Some of the noted standards organizations are

International Standards Organization (ISO)

Institute of Electronics and Electrical Engineers (IEEE)

American National Standards Institute (ANSI)

Ethernet:

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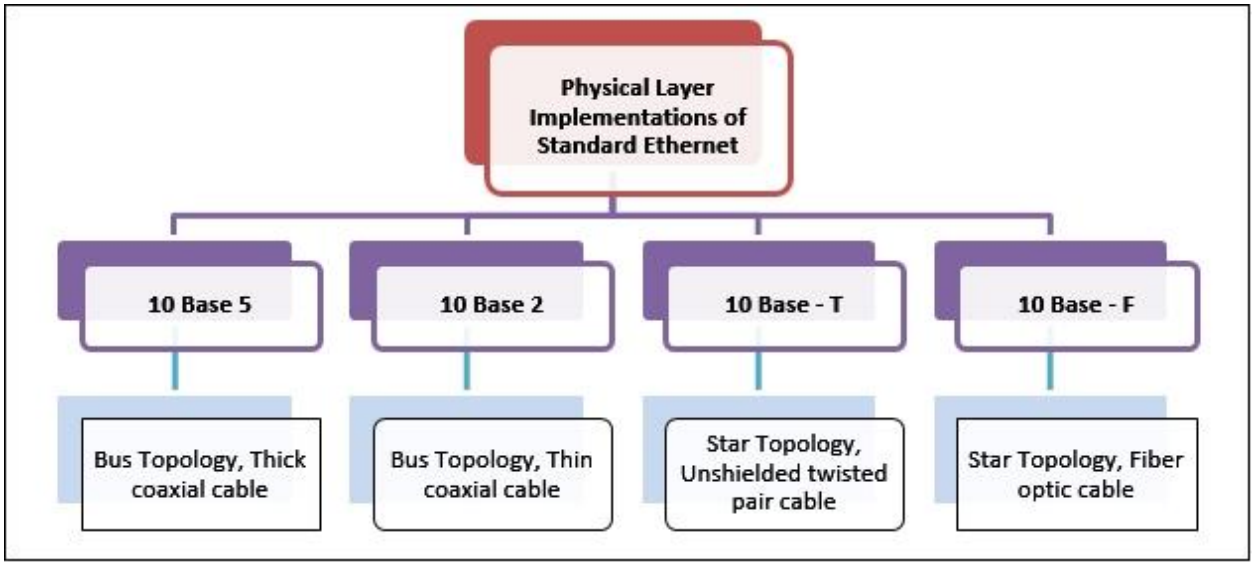
Standard Ethernet has many physical layer implementations. The four main physical layer implementations are shown in the following **diagram**



Ethernet is a network communication protocol designed to link network appliances, switches, and routers. Ethernet functions in a wired or wireless network, including wide area networks (WAN) and local area networks (LAN).

Advancement in Ethernet technology resulted from various network needs such as system application to both large and small platforms, security issues, network dependability, and bandwidth requirements.

Specifically, the demands for higher bandwidth capacity for high-quality video transmission and uninterrupted video streaming brought upon the development of Fast Ethernet and Gigabit Ethernet.

In this piece, we'll explain how the two are different and what you need to consider when selecting an Ethernet data protocol.

Fast Ethernet

Ethernet was standardized in the mid-80s as IEEE 802.3 and continues to develop and improve with dependability and speed. Standard Ethernet speeds averaged at about 10 mbps.

Fast Ethernet is a version of Ethernet standards, initiated in 1995 as IEEE 802.3u. Fast Ethernet focused on increased network and network appliance speed over standard Ethernet and Ethernet devices. Fast Ethernet provided uniform operability for data transmission at over 100 mbps.

It is designed for 100 Base T networks and is also compatible with 10 Base T networks, allowing users to benefit from faster Ethernet speeds (with the use of compatible switches) without having to completely upgrade their network systems.

Fast Ethernet transmission is at least 10 times faster than standard Ethernet and proves to be useful in maintaining compatible connections to high-speed servers, reducing bandwidth bottlenecks for network systems operating multiple IP video cameras and IoT, and seamlessly supporting complex networks that run multiple bandwidth-intensive software packages at the same time.

Fast Ethernet appliances and devices can run on unshielded twisted pair (UTP) cabling such as Category 5 and higher rated cables. However, the maximum length and effectiveness of UTP cabling is 100 meters and bandwidth can be limited.

Fiber optic transmission allows for longer distance and greater bandwidth capacity over UTP cabling, permitting Fast Ethernet network appliances and devices to deliver at full potential.

Gigabit Ethernet

Gigabit Ethernet is the latest version of Ethernet standards, initiated in 1999. Gigabit Ethernet provided uniform standards for 1000 megabits per second, or 1 gigabit per second, network transmission.

Gigabit Ethernet network appliances and devices can transfer data packets at rates much faster than Fast Ethernet. A Fast Ethernet switch can transfer data packets at a rate of 10 mbps . A Gigabit Ethernet switch transfers data packets at relatively higher speeds of 1gbps.

In comparison, a Gigabit Ethernet switch can transfer data packets at around 100 times faster than a Fast Ethernet . Gigabit Ethernet can meet increasingly complex network demands such as connecting multiple bandwidth-intensive devices and broadband internet connections for video streaming.

Gigabit Ethernet applications include gigabit switches that can manage data transfer between multiple IP security cameras and network appliances, and gigabit switches that support video and other high-quality signal transfer between home servers and high-definition televisions and monitor.

10 Base 2

10Base2 is among the family of Ethernet network standards for local area networks (LAN) that uses a thinner version of coaxial cable to establish a network path or medium and operates at a speed of 10 Mbps to carry out baseband transmission.

10Base2 is also known as cheapernet, thinwire, thinnet and thin Ethernet.

10Base2 use coaxial cable that is thinner, flexible, cheaper and comparatively easier to install than cable used in 10Base5 standard. The maximum cable length within a 10Base2 is 200 meters, but 185 meter is a preferred length. The maximum number of nodes that can connect within a 10Base2 network segment is 30.

10Base2 use BNC T-connector to join two cables together and to connect with the computer's network interface card (NIC). Moreover, the cable must be terminated with a 50 ohm resistant at each end.

10 Base 5

10Base5 refers to a standard for Ethernet network technologies that use a thicker version of coaxial cables. It has the capability to transmit data at speed of 10Mbps up to 500 meters using baseband transmission.

10Base5 is also known as Thicknet, ThickWire, thick Ethernet and thick coaxial Ethernet.

The 10Base5 is among the first of original standards of Ethernet along with 10Base2. The name 10Base5 is derived from its 10Mbps transmission speed and 500 meters as the maximum segment length. Its coaxial cable is similar to RG-8/U but with a much thicker shield and braiding, making it highly resistant to network interference. Its outer sheath even has the ability to resist and protect against fire.

A 10Base5 network segment (LAN) can consist of a maximum of 100 nodes. Each node connects to the network segment or cable using an N-connector and must have a distance of 2.5m between one another. Nodes can also be connected through a vampire clam that enables adding a new node in a live network.

10 BASE-T

10BASE-T is an Ethernet standard for local area networks and one of several physical media specified in the Institute of Electrical and Electronics Engineers (IEEE) 802.3 standard. 10BASE-T supports 10 megabits per second (Mbps) transmission speed over twisted-pair cabling with a maximum length of 100 meters (m). The twisted-pair cables connect with an RJ45 connector.

10BASE-T is a shorthand identifier designated by IEEE. The 10 refers to a maximum transmission speed of 10 Mbps. BASE refers to baseband signaling, which means that it can only carry Ethernet signals on the medium. T refers to twisted as in twisted-pair cabling.

In the 1980s, inexpensive twisted-pair wires were widely used to deploy 10BASE-T, which was easier to install than thick and thin coaxial cables. 10BASE-T Ethernet was also the first version of Ethernet to use a star architecture.

Used with a hub, 10BASE-T shares the 10 Mbps bandwidth with different ports. 10BASE-T can also be used with a switch that has full 10 Mbps transmit and receive pairs.

10 BASE-F

10Base-F is a type of standard for implementing Ethernet networks. 10BaseF is different from other 10-Mbps Ethernet technologies because it uses fiber-optic cabling instead of copper unshielded twisted-pair (UTP) cabling. 10BaseF is based on the 802.3 specifications of Project 802 developed by the Institute of Electrical and Electronic Engineers (IEEE).

10BaseF is similar to 10BaseT in that each station is wired into a fiber-optic hub in a star topology to form the network. The maximum length of any segment of 10BaseF fiber-optic cabling is 2 kilometers. The recommended cabling type for 10BaseF networks is 62.5-micron diameter fiber-optic cabling. Two-strand multimode fiber-optic cabling is used, with one strand allotted for transmitting data and the other for receiving data.

What is a network protocol?

A network protocol is a set of established rules that specify how to format, send and receive data so that computer network endpoints, including computers, servers, routers and virtual machines, can communicate despite differences in their underlying infrastructures, designs or standards.

To successfully send and receive information, devices on both sides of a communication exchange must accept and follow protocol conventions. In networking, support for protocols can be built into the software, hardware or both.

What is IP?

Here, IP stands for **internet protocol**. It is a protocol defined in the TCP/IP model used for sending the packets from source to destination. The main task of IP is to deliver the packets from source to the destination based on the IP addresses available in the packet headers. IP defines the packet structure that hides the data which is to be delivered as well as the addressing method that labels the datagram with a source and destination information.

An IP protocol provides the connectionless service, which is accompanied by two transport protocols, i.e., TCP/IP and UDP/IP, so internet protocol is also known as TCP/IP or UDP/IP.

The first version of IP (Internet Protocol) was IPv4. After IPv4, IPv6 came into the market, which has been increasingly used on the public internet since 2006.

History of Internet Protocol

The development of the protocol gets started in 1974 by **Bob Kahn and Vint Cerf**. It is used in conjunction with the Transmission Control Protocol (TCP), so they together named the TCP/IP.

The first major version of the internet protocol was IPv4, which was version 4.

After IPv4, the second major version of the internet protocol was IPv6, which was version 6. It was officially declared by the IETF in 1998. The main reason behind the development of IPv6 was to replace IPv4. There is a big difference between IPv4 and IPv6 is that IPv4 uses 32 bits for addressing, while IPv6 uses 128 bits for addressing.

Function

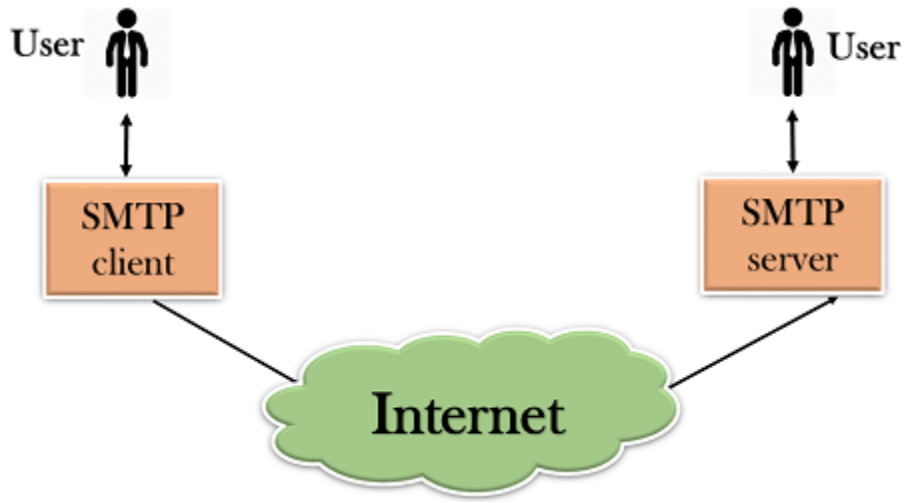
The main function of the internet protocol is to provide addressing to the hosts, encapsulating the data into a packet structure, and routing the data from source to the destination across one or more IP networks.

SMTP Protocol

- SMTP stands for Simple Mail Transfer Protocol.
- SMTP is a set of communication guidelines that allow software to transmit an electronic mail over the internet is called **Simple Mail Transfer Protocol**.
- It is a program used for sending messages to other computer users based on e-mail addresses.
- It provides a mail exchange between users on the same or different computers, and it also supports:
    - It can send a single message to one or more recipients.
    - Sending message can include text, voice, video or graphics.
    - It can also send the messages on networks outside the internet.
- The main purpose of SMTP is used to set up communication rules between servers. The servers have a way of identifying themselves and announcing what kind of communication they are trying to perform. They also have a way of handling the errors such as incorrect email

address. For example, if the recipient address is wrong, then receiving server reply with an error message of some kind.

Components of SMTP



PPP Protocol

The PPP stands for Point-to-Point protocol. It is the most commonly used protocol for point-to-point access. Suppose the user wants to access the internet from the home, the PPP protocol will be used.

It is a data link layer protocol that resides in the layer 2 of the OSI model. It is used to encapsulate the layer 3 protocols and all the information available in the payload in order to be transmitted across the serial links. The PPP protocol can be used on synchronous link like ISDN as well as asynchronous link like dial-up. It is mainly used for the communication between the two devices.

It can be used over many types of physical networks such as serial cable, phone line, trunk line, cellular telephone, fiber optic links such as SONET. As the data link layer protocol is used to identify from where the transmission starts and ends, so ISP (Internet Service Provider) use the PPP protocol to provide the dial-up access to the internet.

Services provided by PPP

- It defines the format of frames through which the transmission occurs.
- It defines the link establishment process. If user establishes a link with a server, then "how this link establishes" is done by the PPP protocol.
- It defines data exchange process, i.e., how data will be exchanged, the rate of the exchange.
- The main feature of the PPP protocol is the encapsulation. It defines how network layer data and information in the payload are encapsulated in the data link frame.
- It defines the authentication process between the two devices. The authentication between the two devices, handshaking and how the password will be exchanged between two devices are decided by the PPP protocol.

FTP Protocol

- FTP stands for File transfer protocol.
- FTP is a standard internet protocol provided by TCP/IP used for transmitting the files from one host to another.
- It is mainly used for transferring the web page files from their creator to the computer that acts as a server for other computers on the internet.
- It is also used for downloading the files to computer from other servers.

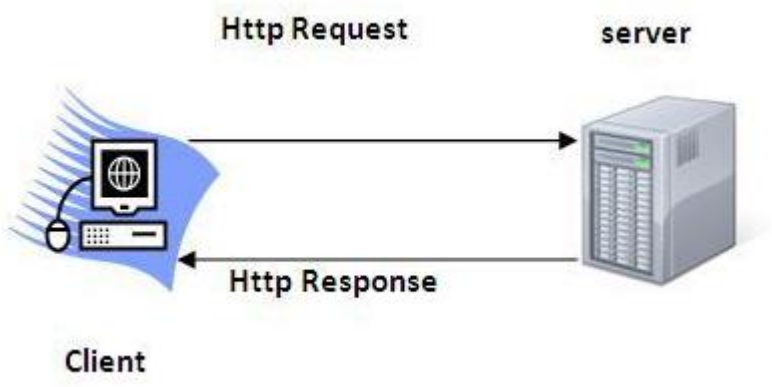
Objectives of FTP

- It provides the sharing of files.
- It is used to encourage the use of remote computers.
- It transfers the data more reliably and efficiently.

HTTP (Hyper Text Transfer Protocol)

The Hypertext Transfer Protocol (HTTP) is application-level protocol for collaborative, distributed, hypermedia information systems. It is the data communication protocol used to establish communication between client and server.

HTTP is TCP/IP based communication protocol, which is used to deliver the data like image files, query results, HTML files etc on the World Wide Web (WWW) with the default port is TCP 80. It provides the standardized way for computers to communicate with each other.



The Basic Characteristics of HTTP (Hyper Text Transfer Protocol):

- It is the protocol that allows web servers and browsers to exchange data over the web.
- It is a request response protocol.
- It uses the reliable TCP connections by default on TCP port 80.
- It is stateless means each request is considered as the new request. In other words, server doesn't recognize the user by default.

- # SNMP Protocol

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- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

SNMP Concept

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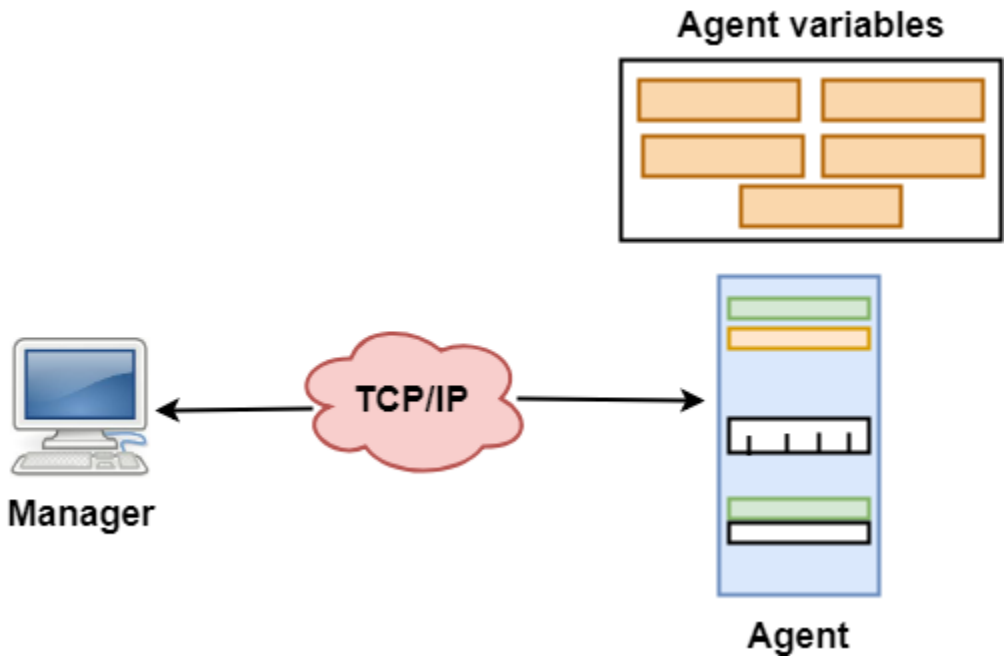

- SNMP has two components Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.

- It is an application layer protocol in which a few manager stations can handle a set of agents.
- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

IP address

- An IP address represents a unique address that distinguishes any device on the internet or any network from another. IP or Internet Protocol defines the set of commands directing the setup of data transferred through the internet or any other local network.
- An IP address is the identifier that enables your device to send or receive data packets across the internet. It holds information related to your location and therefore making devices available for two-way communication. The internet requires a process to distinguish between different networks, routers, and websites. Therefore, IP addresses provide the mechanism of doing so, and it forms an indispensable part in the working of the internet. You will notice that most of the IP addresses are essentially numerical. Still, as the world is witnessing a colossal growth of network users, the network developers had to add letters and some addresses as internet usage grows.
- An IP address is represented by a series of numbers segregated by periods(.). They are expressed in the form of four pairs - an example address might be 255.255.255.255 wherein each set can range from 0 to 255

Sliding Window Protocol

The sliding window is a technique for sending multiple frames at a time. It controls the data packets between the two devices where reliable and gradual delivery of data frames is needed. It is also used in TCP (Transmission Control Protocol).

In this technique, each frame has sent from the sequence number. The sequence numbers are used to find the missing data in the receiver end. The purpose of the sliding window technique is to avoid duplicate data, so it uses the sequence number.

Pipelining

The term Pipelining refers to a technique of decomposing a sequential process into sub-operations, with each sub-operation being executed in a dedicated segment that operates concurrently with all other segments.

The most important characteristic of a pipeline technique is that several computations can be in progress in distinct segments at the same time. The overlapping of computation is made possible by associating a register with each segment in the pipeline. The registers provide isolation between each segment so that each can operate on distinct data simultaneously.

The structure of a pipeline organization can be represented simply by including an input register for each segment followed by a combinational circuit.

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